OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

14.2 Lane validity for signals

By default, signals are valid for all lanes of a road, for traffic traveling in the direction indicated by @orientation attribute of a `<signal>` element. Lane validity offers the possibility to restrict the validity of a signal to specific lanes only by using a `<validity>` element.

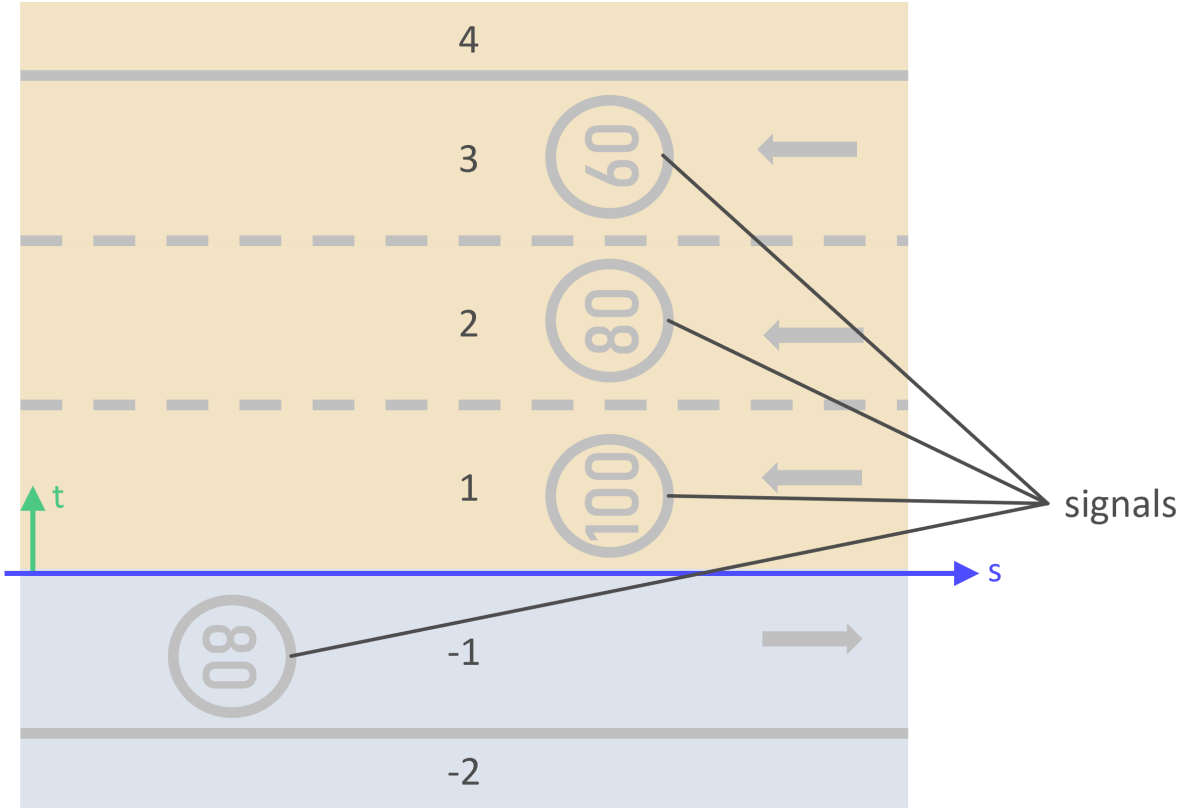

Figure 128. Lanes with signals in the shape of road marks

Figure 128 shows how signals in the shape of a road mark specify the speed limit of different lanes.

NOTE

NOTE

The @orientation attribute and `<validity>` element complement each other. The @orientation attribute and the `<validity>` element are not interchangeable.

- The @orientation attribute defines the travel direction for which a signal is valid.
- The `<validity>` element defines the lanes for which a signal is valid.

As an example for the difference in using the attribute and the element, speed limits can be taken: if traveling in road reference line direction, with right-hand-traffic, then a speed limit signal with `orientation="+"` applies to a vehicle even if this vehicle is driving on an oncoming lane while overtaking. If the validity is limited to all right lanes then the signal does not apply, however, to the vehicle while it is in an oncoming lane. Therefore, the `<validity>` element should only be used to limit signals to specific lanes, for example for traffic lights which only apply to certain lanes.

Elements in UML model

`<validity>` element

In ASAM OpenDRIVE, lane validity is represented by the `<validity>` element within the `<signal>` or `<signalReference>` element.

```
UML class: t_road_objects_object_laneValidity
XML tag:   <validity> (Multiplicity: 0..*)
```

Lane validities restrict signals and objects to specific lanes.

Table 121. Attributes of the <validity> element

Name	Type	Use	Description
`fromLane`	integer	required	Minimum ID of the lanes for which the object is valid
`toLane`	integer	required	Maximum ID of the lanes for which the object is valid

Rules

The following rules apply to validity elements:

- A signal may be valid for one or more lanes.
- The range given by all `<validity>` elements shall be a subset of the parent's @orientation attribute:
- asam.net:xodr:1.7.0:road.signal.validty.right_hand_traffic_lane_ids: For right-hand traffic, @orientation="+" implies that the `<validity>` element shall only span negative lane ids, while @orientation="-" implies that the `<validity>` element shall only span positive lane ids. If the given `<validity>` elements span both, positive and negative lane ids, @orientation="none" shall be used.
- asam.net:xodr:1.7.0:road.signal.validty.left_hand_traffic_lane_ids: For left-hand-traffic, @orientation="-" implies that the `<validity>` element shall only span negative lane ids, while @orientation="+" implies that the `<validity>` element shall only span positive lane ids. If the given `<validity>` elements span both, positive and negative lane ids, @orientation="none" shall be used.
- The value of the @fromLane attribute shall be lower than or equal to the value of the @toLane attribute.

Related topics

- Section 14.1, "Introduction to signals"
- Section 14.3, "Signal dependency"